

flows from the M side to the N side, B moves in the opposite direction, and only the loaded carrier can cross the membrane. Here, distinct affinities and dissociation constants at the two sides of the membrane and for each of the solutes are considered, and the forward and backward translocation rate constants of a given solute can differ. In this model, the rapid binding and unbinding assumption and the quasi-steady-state assumption are used. Under these assumptions, the outward fluxes for each of the solutes A , B , and C are obtained by equations 64a, 64b, and 64c, respectively [100].

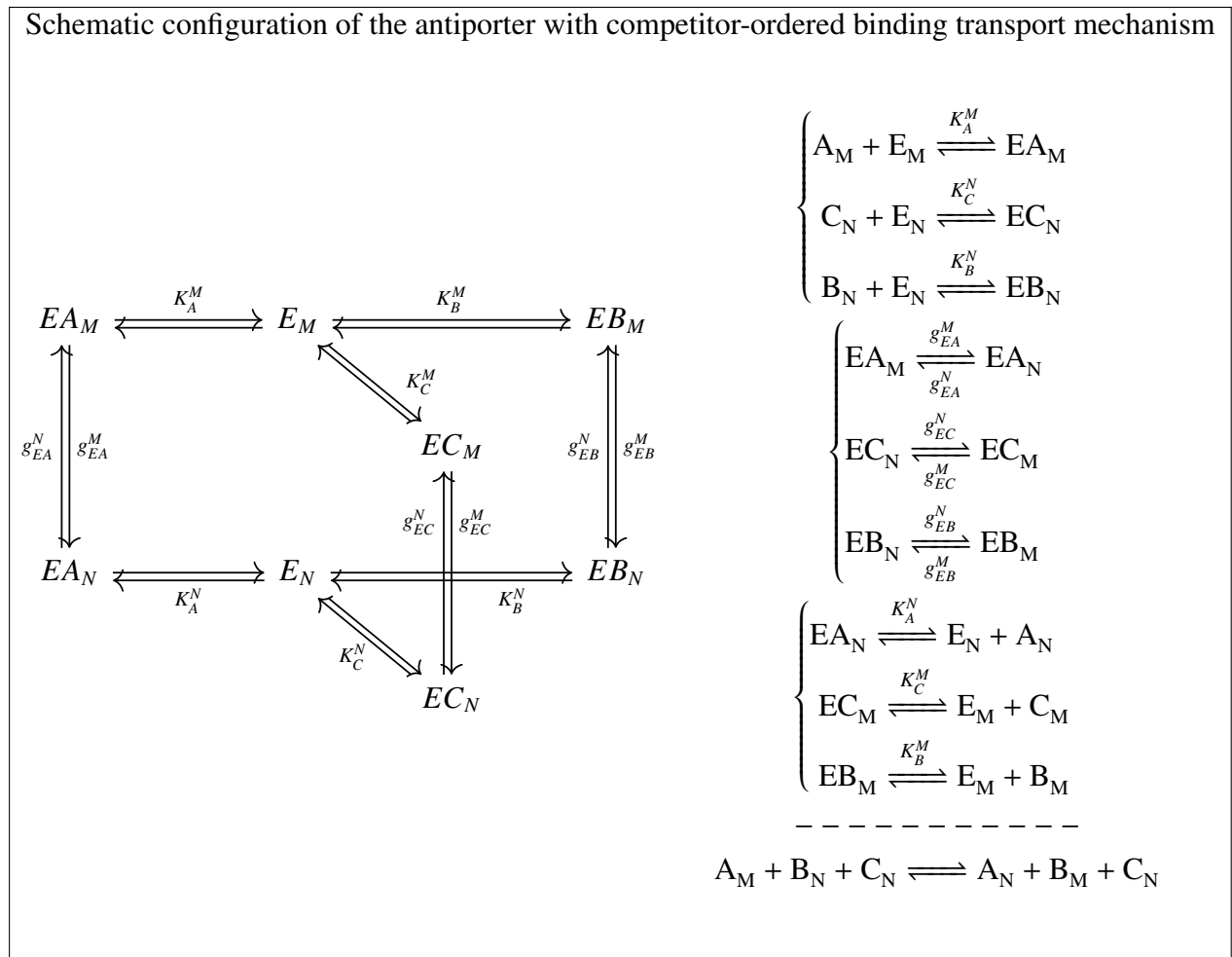


Figure 21: A schematic configuration of the antiporter with competitor-ordered binding transport mechanism of two solutes and the regarding net transport flux equations. In this configuration, the carrier exists in four different states on each side of the membrane. Transport equations are reparameterized to the general model from Layton model [100]. Dissociation constants (K_A , K_B , and K_C terms), and the translocation rate constants (g_{EA} , g_{EB} , and g_{EC} terms) are depicted in the reaction cycle in the right panel. Subscripts M and N, denote the M and N sides of the membrane. $[E]_t$ is the total number of transporter per unit area of the membrane. (part 1/2 continued on next page).

Net Transport Fluxes	
$J_{A,antiporter}^{M,N(net)} = [E]_t \frac{g_{EA}^M \alpha^M (g_{EB}^N \beta^N + g_{EC}^N \gamma^N) - g_{EA}^N \alpha^N (g_{EB}^M \beta^M + g_{EC}^M \gamma^M)}{R_M R_{NN} + R_N R_{MM}}$	(64a)
$J_{B,antiporter}^{M,N(net)} = [E]_t \frac{g_{EB}^M \beta^M (g_{EA}^N \alpha^N + g_{EC}^N \gamma^N) - g_{EB}^N \beta^N (g_{EA}^M \alpha^M + g_{EC}^M \gamma^M)}{R_M R_{NN} + R_N R_{MM}}$	(64b)
$J_{C,antiporter}^{M,N(net)} = [E]_t \frac{g_{EC}^M \gamma^M (g_{EA}^N \alpha^N + g_{EB}^N \beta^N) - g_{EC}^N \gamma^N (g_{EA}^M \alpha^M + g_{EB}^M \beta^M)}{R_M R_{NN} + R_N R_{MM}}$	(64c)

where

$$[E]_t = [E]_M + [EA]_M + [EB]_M + [EC]_M + [E]_N + [EA]_N + [EB]_N + [EC]_N$$

$$\alpha^M = \frac{[A]_M}{K_A^M}, \quad \beta^M = \frac{[B]_M}{K_B^M}, \quad \gamma^M = \frac{[C]_M}{K_C^M}$$

$$R_M = 1 + \alpha^M + \beta^M + \gamma^M$$

$$R_{MM} = g_{EA}^M \alpha^M + g_{EB}^M \beta^M + g_{EC}^M \gamma^M$$

$$\alpha^N = \frac{[A]_N}{K_A^N}, \quad \beta^N = \frac{[B]_N}{K_B^N}, \quad \gamma^N = \frac{[C]_N}{K_C^N}$$

$$R_N = 1 + \alpha^N + \beta^N + \gamma^N$$

$$R_{NN} = g_{EA}^N \alpha^N + g_{EB}^N \beta^N + g_{EC}^N \gamma^N$$

Figure 21: A schematic configuration of the antiporter with competitor-ordered binding transport mechanism of two solutes and the regarding net transport flux equations. In this configuration, the carrier exists in four different states on each side of the membrane. Transport equations are reparametrized and transformed into the general resistance form from Layton model [100]. Dissociation constants (K_A , K_B , and K_C terms), and the translocation rate constants (g_{EA} , g_{EB} , and g_{EC} terms) are depicted in the reaction cycle in the right panel. Subscripts M and N, denote the M and N sides of the membrane. $[E]_t$ is the total number of transporter per unit area of the membrane (part 2/2 continued from previous page).

Antiporter simplified model:

Simplified models of antiport and symport transporters have been developed using similar assumptions. Sohma et al. (1996) developed a simplified antiporter model for the one-to-one transport of two substrates [138]. Here, their model is extended to the general case of an antiporter transporting three solutes in a ratio of $aA : bB : cC$. The following assumptions are used: 1) the rapid binding and unbinding assumption, 2) the quasi-steady-state assumption, and 3) the symmetric carrier assumption (i.e., the velocity constants $g_{antiporter}$ for transport from the M to the N side of the membrane vice-versa are the same, $g_{EA}^M = g_{EA}^N = g_{antiporter}$, and the dissociation constants for each solute on both faces are the same, $K_A^M = K_A^N = K_A$, $K_B^M = K_B^N = K_B$), 4) the antiporter has a single binding site for the transported solutes, 5) the transported solutes compete for the same binding sites, and 6) finally, the empty antiporter does not cross the membrane. Using these as-